

CRITICAL

Account Takeover

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Routed to: Compliance Team · Fraud Risk Team · Tm Analytics Team · Policy Controls Team

EXECUTIVE SUMMARY

A CRITICAL priority Account Takeover threat has been identified affecting UOB's RETAIL customer segment. Approximately 180,000 customers are estimated at risk across UNKNOWN, SG. Current transaction monitoring controls leave a significant detection gap, requiring immediate attention. Immediate action: issue customer advisory and escalate to Fraud Risk for TM rule deployment.

THREAT INTELLIGENCE BRIEF

Account Takeover attacks exploit Malware-as-a-Service (MaaS) distributed via Telegram/Social Media to compromise customer accounts and execute unauthorized transactions. Android Accessibility malware disguised as a local utility application intercepting SMS OTPs. This typology was first reported in UNKNOWN, SG and has been escalating across the region.

UOB RELEVANCE ASSESSMENT

Criteria	Result	Details
Criterion 1 — Novelty	✓ PASSED	No existing TM rule covers this typology; Similar HIGH/CRITICAL signal alerted within last 30 days (8 found); High novelty
Criterion 2 — Customer Exposure	✓ PASSED	✓ RETAIL customers in UNKNOWN/SG; ~180,000 at risk
Criterion 3 — Control Gap	✓ PASSED	✓ 100% of attacks undetected by existing TM rules; auth vulnerability: YES

Composite Risk Score: 98/100 — CRITICAL Priority

UOB's RETAIL customer base in Singapore and the ASEAN region is directly targeted by this fraud pattern. The bank's current authentication methods and transaction monitoring coverage leave approximately 100% of attacks undetected. With a composite risk score of 98/100, this signal is classified as CRITICAL.

FINANCIAL IMPACT SIMULATION

Scenario	Exposure (SGD)
Baseline (no intervention)	SGD 980K
With current TM controls	SGD 980K
With proposed interventions	SGD 431K
P10 (optimistic)	SGD 1.5M
P50 (median)	SGD 4.7M
P90 (worst case)	SGD 15.1M

Baseline exposure (no intervention): SGD 980K. With current controls: SGD 980K. With proposed interventions: SGD 431K. Monte Carlo range (P10–P90): SGD 1.5M – SGD 15.1M.

RECOMMENDED ACTIONS

■ Immediate (<24 hours)

- Issue advisory push notification to at-risk RETAIL customers
- Activate enhanced monitoring on new payee transfers
- Alert fraud operations team for manual review queue

■ Short-term (1–7 days)

- Submit ACCOUNT_TAKEOVER TM detection scenario to analytics team
- Increase step-up authentication threshold for high-risk transactions
- Update customer-facing fraud guidance on website and app
- Brief relationship managers on identification and escalation procedure

■ Strategic (1–4 weeks)

- Accelerate device binding rollout to reduce SMS OTP dependency
 - Commission technical assessment of real-time behavioral analytics capability
 - Develop full response playbook for Account Takeover incidents
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PROPOSED INTERVENTION RULES

1. TM_DETECTION Risk: MEDIUM · Impact: ~25% · Auto-deployable: No

Flags suspicious transactions originating from a newly registered device, especially when authenticated via SMS OTP and involving high-value transfers or new beneficiaries.

```
IF (Successful_Login_From_New_Device AND Auth_Method == 'SMS_OTP' AND (Transaction_Type == 'Fund_Transfer_New_Beneficiary' OR Transaction_Amount > 5000 OR Change_Contact_Details_Attempt)) THEN Generate_Alert_For_Manual_Review
```

■ Requires Head of Fraud Risk sign-off

2. CUSTOMER_FRICTION Risk: MEDIUM · Impact: ~35% · Auto-deployable: No

Implements an additional authentication challenge for high-risk transactions initiated from Android devices, requiring confirmation via UOB Mighty Secure push notification instead of SMS OTP.

```
IF (Device_OS == 'Android' AND Transaction_Risk_Score > 70 AND Auth_Method_Requested == 'SMS_OTP') THEN Force_UOB_Mighty_Secure_Push_Auth_Challenge
```

■ Requires Head of Fraud Risk sign-off

3. POLICY_CHANGE Risk: HIGH · Impact: ~40% · Auto-deployable: No

Mandate the use of UOB Mighty Secure (push notification/biometric) for all high-risk transactions and device registrations, deprecating SMS OTP for these specific actions.

```
UPDATE Policy: For (High_Risk_Transactions OR New_Device_Registration), REQUIRE Auth_Method == 'UOB_Mighty_Secure'; DISALLOW Auth_Method == 'SMS_OTP'
```

■ Requires Head of Fraud Risk sign-off

4. ADVISORY Risk: LOW · Impact: ~15% · Auto-deployable: Yes

Distribute an advisory to all retail customers, especially Android users, warning about Android Accessibility malware disguised as utility apps and advising on safe app downloads and UOB Mighty Secure usage.

```
SEND_EMAIL_SMS_PUSH_NOTIFICATION_TO_RETAIL_CUSTOMERS_WITH_MESSAGE('Beware of Android Accessibility malware disguised as utility apps. Only download apps from official stores. Enable UOB Mighty Secure for stronger protection.')
```

■ Requires Head of Fraud Risk sign-off

EVIDENCE SOURCES

- 1. Intelligence derived from regulatory advisory and open-source threat data.

ANALYST NOTES & CAVEATS

This alert is AI-generated. It must be reviewed by a qualified fraud risk analyst before operational action is taken. IMPORTANT: This alert is AI-generated using synthetic/mock data for POC purposes only. It must be reviewed by a qualified fraud risk analyst before operational action is taken. Data confidence: 90%. All financial figures are estimates derived from synthetic customer population data. No real UOB customer data was used in this analysis.